

Requirement Analysis

Solution Requirements (Functional & Non-functional)

Date	14-07-2026
Team ID	
Project Name	EduGenie
Maximum Marks	

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Intelligent Question Answering	Accept a natural-language question via the input box Return an accurate answer with brief educational context Handle follow-up/contextual questions within a session
FR-2	Simplified Concept Explanations	Accept a topic/concept name from the learner Return a plain-language explanation suited to the learner's level
FR-3	AI-Powered Quiz Generation	Accept a topic and generate topic-specific quiz questions Support self-evaluation by revealing answers after attempt Vary question difficulty across a quiz set
FR-4	Educational Text Summarization	Accept pasted/uploaded learning material Return a concise, well-organized summary
FR-5	Personalized Learning Path Recommendations	Accept a subject of interest from the learner Generate a staged roadmap: beginner, intermediate, advanced Include study recommendations per stage
FR-6	Interactive, User-Friendly Interface	Provide a single responsive interface for all features Show loading indicators while AI responses are generated

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The interface must be simple and intuitive for students, self-learners, and educators of varying technical comfort
NFR-2	Security	API keys and credentials for the generative AI service must be stored securely and never exposed to the frontend
NFR-3	Reliability	The system should handle malformed or ambiguous input gracefully without crashing
NFR-4	Performance	Responses should return quickly enough to feel interactive and conversational

NFR-5	Availability	The core Q&A, explanation, quiz, summarization, and learning path features should be accessible whenever the app is running
NFR-6	Scalability	The modular backend architecture should allow additional features or model integrations to be added with minimal rework
NFR-7	Portability / Lightweight Footprint	The system must run efficiently on resource-constrained devices such as the Mac M1